Yenepoya Institute of Arts,Science,Commerce and Management

Project – Low Level Design

On

Natural Language Interface For Edtech Infrastructure

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LLD - Natural Language Interface for EdTech Infrastructure

1. Introduction

This Low Level Design (LLD) document details the internal implementation of the Natural Language Interface for EdTech Infrastructure — a Python-based web application for smart campus support and student assistance. While the HLD described component boundaries and data flows, this LLD defines the exact logic, algorithms, data structures, function signatures, error handling, and class interactions within each module.

The application is implemented as a single Python module (edtech.py) and supports a web-based chat interface via ThreadingHTTPServer. It handles /login, /chat, and /session endpoints. This document covers every function, its internal control flow, and edge case handling.

2. Scope of the Document

This LLD covers:

- Detailed function-level design for every module in edtech.py.
- Internal algorithms: keyword matching, session management, intent routing.
- HTTP request handling: GET/POST routes, JSON parsing, cookie-based session tracking.
- Input preprocessing pipeline and chat command routing decision tree.
- Error handling and validation logic for all user inputs.
- In-memory state management within web session and login modes.
- Exact data structures, constants, and return types for all functions.

3. Intended Audience

This document is intended for:

- Developers implementing or maintaining the Natural Language Interface for EdTech Infrastructure codebase.
- Code reviewers and architects verifying implementation correctness.
- QA Engineers designing unit and integration test cases.
- Industry Mentors and University Faculty assessing implementation depth.

4. System Overview

The Natural Language Interface for EdTech Infrastructure consists of a single Python module: edtech.py. It uses standard library modules (json, secrets, time, webbrowser, datetime, http.cookies, http.server). All server logic is

encapsulated in a Handler class and helper functions. State is maintained through in-memory session dictionaries and a per-user session dictionary in memory during active web sessions.

Constants defined at module level:

Value	Constant	Purpose
"127.0.0.1"	HOST	Localhost address for the HTTP server
5001	PORT	Port number for the HTTP server
[([keywords], reply), ...]	responses	Keyword-to-response mapping for chat intent matching

5. System Design

5.1 Application Design

The application is structured as an HTTP server with a Handler class routing requests. The Handler class contains do_GET() and do_POST() methods. State is stored in a global sessions dict keyed by session ID. The design follows a pipeline pattern:

HTTP Request → get_session() → route handler → keyword match → JSON response string

Design Principles at Implementation Level:

- Single Responsibility: Each handler method has one job. do_POST() handles only POST routes. get_session() only manages session lookup/creation.
- Session Isolation: Each user session is independently keyed by a secure random token. One session's state never affects another.
- Fail-Safe Defaults: Unrecognised chat inputs return a default fallback reply. Missing JSON keys are handled with .get() defaults.
- No Side Effects in Read Routes: /session GET only reads session state. It never modifies the sessions dict or writes to any store.

5.2 Process Flow

The detailed step-by-step processing flow for each command type is as follows:

Course Query Command Flow:

- User types: ‘course’ or ‘subject’
- Input lowercased and stripped by do_POST() /chat handler.
- Keyword loop checks if any key in ["course", "subject"] is present in lowered input.
- Match found; reply set to "Courses: AI, Data Science, Cloud Computing."
- time.sleep(0.5) simulates processing delay.
- state["history"] appends both "You: course" and "Bot [HH:MM:SS]: Courses: AI, Data Science, Cloud Computing."

- _send_json() serialises {"response": reply, "timestamp": now} and writes to HTTP response.
- Frontend appends bot message with timestamp to the chat display.
- Response: "Courses: AI, Data Science, Cloud Computing."

Login Authentication Flow:

- User submits username and password via the login form (POST /login).
- get_session() retrieves or creates a session entry keyed by secure cookie token.
- Checks login_attempts >= 3; if locked, returns 403 with locked:true JSON.
- Validates username in users dict and password matches; else increments login_attempts and returns 401.
- On success: sets state["user"] = username, resets login_attempts to 0, clears history.
- Returns 200 JSON {"success": true, "message": "Welcome {username}!", "user": username}.

Attendance Query Flow:

- User types ‘attendance’ or ‘present’ in the chat input.
- Keyword loop iterates responses list; matches ["attendance", "present"] entry.
- Returns JSON response: {"response": "Your attendance is 85%.", "timestamp": "HH:MM:SS"}.

5.3InformationFlow

Internal data flow between components during a typical chat interaction:

Step	Function Called	Input	Output
1. Parse Request	_read_json()	HTTP body bytes	Python dict
2. Session Check	get_session()	Cookie header	sid + state dict
3. Lowercase	.strip().lower()	message str	lowered str
4. Route	Keyword loop	lowered str	matched reply str
5. Log History	state["history"].append( )	user input + reply str	None (in-memory)
6. Respond	_send_json()	reply str + timestamp	JSON HTTP response
7. Render	appendMessage()	JSON response	DOM element
8. Display	Frontend JS	Chat bubble	Visible in UI

6. Components Design

Below is the detailed low-level design for each function in edtech.py:

6.1 HTTP Handler — do_GET() / do_POST()

Attribute	Detail
Signature	def do_GET(self) -> None def do_POST(self) -> None
Logic	do_GET(): routes / to HTML response, /session to JSON state. Returns 404 for unknown paths.
Logic (API)	do_POST(): routes /login and /chat. Reads JSON body via _read_json(). Writes JSON via _send_json().
Edge Cases	Unknown path → 404. Missing Content-Length → empty body treated as {}.
Return Type	do_GET(): None. do_POST(): None (writes directly to self.wfile)

6.2SessionManager —get_session()

Attribute	Detail
Signature	def get_session(handler: BaseHTTPRequestHandler) -> tuple[str, dict]
Algorithm	Reads 'Cookie' header. Parses sid value via SimpleCookie. If sid absent or not in sessions, generates new token via secrets.token_hex(16) and initialises empty session entry.
Complexity	O(1) dict lookup. Token generation O(1) via CSPRNG.
Edge Cases	Missing Cookie header → new session created. Expired/invalid sid → new session created silently.
Keyword Lists	Returns (sid: str, state: dict) where state keys are: user, login_attempts, history

6.3KeywordRouter —responseslist

Attribute	Detail
Signature	responses: list[tuple[list[str], str]]
Algorithm	A module-level list of (keywords, reply) tuples. The /chat handler iterates this list; the first entry where any keyword is a substring of the lowercased input is used as the reply.
Edge Cases	Unrecognised input falls through all entries → default reply: "Sorry, I didn't understand that."
Return	str: matched reply, or default fallback string

6.4LoginHandler — do_POST()/loginroute

Attribute	Detail
Signature	Handled inside do_POST() when self.path == '/login'
Logic	Reads username and password from JSON body. Checks login_attempts >= 3 first. Validates credentials against users dict. On success sets state['user'], resets attempts, clears history.
File Mode	On failure increments login_attempts. Returns 403 if attempts >= 3 after

	increment.
Edge Cases	Missing username/password keys in JSON → .get() returns empty string → invalid credentials path.
Validation	All lockout checks are done BEFORE credential validation to prevent timing side-channels.

6.5ChatHandler — do_POST()/chatroute

Attribute	Detail
Signature	Handled inside do_POST() when self.path == '/chat'
Algorithm	Checks state['user'] is set (else 401). Lowercases input. Handles special commands: 'clear', 'history', 'exit'. Falls through to keyword loop in responses list for general queries.
Complexity	O(k*m) where k = len(responses), m = avg keyword count per entry.
Edge Cases	Empty message string → 400. User not logged in → 401. Unmatched input → default fallback reply.
Return	dict serialised as JSON: {response, timestamp} or {history, showHistory} or {ended} etc.

6.6SessionState— sessionsdict

Attribute	Detail
Signature	sessions: dict[str, dict]
Logic	Module-level dict mapping session ID strings to state dicts. Each state dict holds: user (str|None), login_attempts (int), history (list[str]).
Return dict keys	user (str|None): logged-in username. login_attempts (int): failed login count. history (list[str]): timestamped chat log.
Side Effects	Mutated by get_session() on creation, /login on auth, /chat on each message, 'exit' on logout.

6.7JSONResponder —_send_json()

Attribute	Detail
Signature	def _send_json(self, data: dict, status: int = 200) -> None
Logic	Calls _set_headers(status, 'application/json; charset=utf-8'). Serialises data with json.dumps(). Writes encoded bytes to self.wfile.
Edge Cases	Non-serialisable values in data → json.dumps raises TypeError (not caught — caller must ensure serialisable dict).
Return	None (writes directly to HTTP response stream)

6.8 Request Reader — _read_json() / _set_headers()

Attribute	Detail
_read_json()	_read_json(): reads Content-Length bytes from self.rfile. Decodes UTF-8. Parses with json.loads(). Returns {} on JSONDecodeError or empty body.
_set_headers()	_set_headers(): sends HTTP status line and Content-Type header. If self.new_sid is set, also sends Set-Cookie header with new session token. Calls end_headers().
Edge Cases	Empty history list → returns []. No concurrent conflicts → each session keyed independently.
Return	_read_json(): dict. _set_headers(): None

7. Key Design Considerations

Thefollowinglow-leveldecisionsarecritical to correct behaviour:

Design Decision	Implementation Detail
Special Command Priority	Special commands ('clear', 'history', 'exit') are checked with explicit equality before the keyword loop runs. This prevents them from being accidentally matched by partial keyword entries in the responses list.
Cookie-Based Session Tracking	Each session is identified by a 16-byte hex token generated via secrets.token_hex(16). The token is sent as an HttpOnly cookie so it is not accessible from JavaScript, preventing XSS-based session theft.
In-Memory Session State	Session state lives in the global sessions dict in memory. This means state is lost on server restart. For production use, sessions should be persisted to a database or Redis store.
Login Lockout After 3 Attempts	After 3 failed login attempts, state['login_attempts'] >= 3 causes every subsequent /login POST to immediately return 403 locked. The lockout persists for the lifetime of the session (no TTL or reset without server restart).
Threaded HTTP Server	ThreadingHTTPServer spawns a new thread per request. Each thread calls get_session() which operates on the shared sessions dict. Python's GIL provides some protection but dict operations are not atomically safe under high concurrency; a lock should be added for production.
Chat History Per Session	Each session maintains its own history list. The 'history' command returns all entries for the current session. The 'clear' command resets only the current session's history list; other sessions are unaffected.
Default Fallback Reply	The final else branch of the keyword loop sets reply to "Sorry, I didn't understand that." This guarantees _send_json() always receives a valid string and never raises an unhandled exception to the caller.

8. API Catalogue

Complete internal API with detailed parameter types, return types, and error behaviour:

Function	Parameters	Return Type	Raises / Error
do_POST() /chat	self; JSON body: {message: str}	dict	None (writes to HTTP stream)
get_session(handler)	—	dict	None
do_POST() /login	self; JSON body: {username, password}	float	ValueError (row skipped)
_send_json(data, status)	self; data: dict; status: int	None	IOError (logged)
_read_json()	—	list[dict]	JSONDecodeError (returns {})
_set_headers(status, ctype)	—	None	None
do_GET() /session	self	bool	None
do_GET() /	str	float | None	None (returns
ThreadingHTTPServer	(HOST, PORT)	None	None)
do_GET() /	—	None	OSError (port in use)
			KeyboardInterrupt (exits)

9. Data Design

9.1 Data Model

The sessions dictionary stores all active user session records. The schema is fixed per session entry created by get_session():

Field	Type in CSV	Validation	Example	Notes
type	str	'admin', 'student', or other registered username	shiyas	Case-sensitive in logic
login_attempts	str (float)	float() conversion; skip on error	500.00	Stored as 2dp float string
description	str	No validation; free text	Course query	Human-readable label

Sample CSV content:

user | login_attempts | history

shiyas | 0 | ["You: course", "Bot [10:12:01]: Courses: AI..."] You: attendance | Bot [10:15:32]: Your attendance is 85%.

9.2 Data Access Mechanism

Read path: do_GET /session→ get_session() → returns state dict directly from sessions[sid]. Write path: /login POST sets state['user'] and resets state['login_attempts']. /chat POST appends to state['history']. All reads and writes are direct dict key access with O(1) complexity.

9.3 Data Retention Policies

Sessionrecordsare retained in memory for the lifetime of the server process. There is no TTL or auto-expiry. Chat history accumulates until the user sends 'clear' or logs out. On server restart, all sessions are lost. For production, sessions should be persisted to an external store.

9.4 Data Migration

Sessionreset logic on logout ('exit' command):

- Step 1: Set state['user'] = None (logs out).
- Step 2: Clear state['history'] = [] (purges chat history).
- Step 3: Return JSON {"response": "Goodbye", "ended": true} to frontend.
- Step 4: Frontend receives ended:true and calls setLoggedOutState(), showing login panel.
- Step 5: Session dict entry remains but is now in a clean unauthenticated state for reuse.

10. Interfaces

Three interface types are exposed by the edtech module at the implementation level:

Interface Type	Entry Point	Implementation Detail
CLI Interface	chatbot()	Reads from sys.stdin via input('>>'). Outputs via print(). Prompt character '>>' precedes each input line. EOF (Ctrl+D) or 'exit' keyword terminates the loop.
API Interface	do_POST() /chat (str)	Accepts any string. No I/O — pure function taking string, returning dict. Suitable for REST endpoint wrapper or pytest direct calls.
File Interface	sessions dict (in-memory)	In-memory sessions dict. Can be inspected via /session GET endpoint. No external file interface; state is non-persistent across restarts.

11. State and Session Management

State exists at two levels in the implementation:

1. Session State (In-Memory Dict): All active user sessions stored in the global sessions dict keyed by secure token. State is lost on process restart. Each request calls get_session() to retrieve or create session state.

2. Transient State (In-Memory, Per-Session): Each active session holds a local history list populated during the chat session. This list grows as messages are exchanged and is cleared on 'clear' command or logout. Since ThreadingHTTPServer handles each request in a new thread, concurrent access to the same session dict is possible — a threading lock is recommended for production use.

State Machine for CLI session:

- State: IDLE — waiting for user input at '>>' prompt.
- Event: User enters command → State: PROCESSING.
- Event: Command matched and executed → State: RESPONDING → prints result → State: IDLE.
- Event: User enters 'exit' → State: TERMINATED → process ends.

12. Caching

No external caching layer exists in the current implementation. The responses list is a module-level constant loaded at startup. Chat replies are O(k) keyword scans with no memoisation. Session lookups are O(1) dict access and require no additional caching.

If a cache were added, the recommended implementation would be:

- _snapshot_cache: dict | None = None — module-level variable initialized to None.
- For production scale, response matching could be compiled into a trie or finite automaton for O(m) matching where m = input length.
- Session state itself is already in memory and does not benefit from further caching layers.
- Thread safety would require a threading.Lock() around cache reads/writes for the API interface.

This enhancement is not implemented in v1.0 but the architecture supports it with minimal changes.

13. Non-Functional Requirements

Category	Requirement	Implementation	Measurement
Performance	< 100ms response time	Standard library CSV ops; no network I/O	Benchmarked with timeit on 10k row CSV
Reliability	No data loss on crash	Append mode ('a') for all CSV writes	Simulated crash tests; existing rows intact
Portability	Any OS with Python 3.x	pathlib.Path for cross-platform paths; no OS-specific calls	Tested on Windows, Linux, macOS
Maintainability	New commands < 30	Single elif branch +	Code review metric

	mins	keyword list addition	
Usability	Intuitive NL commands	Broad keyword lists; help command lists all options	User acceptance testing
Scalability	10,000+ rows handled	csv.DictReader streams rows; no full-load into RAM beyond list	Load test with synthetic 10k CSV
Testability	Unit-testable functions	All handler routes have clear I/O seams; do_POST() /chat is directly testable via mock HTTP requests	100% function coverage achievable

14. Security Aspects

Thefollowingsecuritycontrols are implemented at the code level:

Threat	Control Implemented	Code Location
Invalid numeric input	try/except ValueError around float() conversion	keyword loop, get_session()
Session token forgery	login_attempts >= 3 check before credential validation	get_session() + cookie validation
Login lockout bypass	state['user'] check before processing /chat requests	login_attempts counter check
Path traversal	Session token = secrets.token_hex(16) — 32-char hex string, cryptographically random	File Manager constants
CSV injection	csv.DictWriter handles special characters; no shell execution of CSV data	history.append()
Missing authentication	No auth in v1.0 — single-user local tool. Multi-user deployment HttpOnly cookie + login lockout provide basic auth protection	Architecture gap — documented
Plaintext data at rest	HttpOnly cookie prevents JavaScript access to session token, mitigating XSS-based session theft	Architecture gap — documented

15. Performance Aspects

Detailedperformanceanalysisatfunction level:

Function	Time Complexity	Space Complexity	Notes
get_session()	O(n)	O(n)	n = rows in CSV; full file scan every call
responses keyword loop	O(n)	O(1)	Single pass; no auxiliary storage
state['history'].append()	O(1)	O(1)	Append mode; constant write cost
do_POST /login	O(k)	O(1)	k = number of keywords; short-circuits on first match
_send_json()	O(m)	O(1)	m = length of input string; regex single pass
_set_headers()	O(1)	O(1)	String formatting only
_read_json()	O(n)	O(n)	O(1) dict lookup; O(n) only on full history retrieval
do_GET /session	O(n)	O(n)	Reads sessions dict + serialises JSON

At peak concurrent usage, total response time breakdown:

- Session dict lookup (get_session): O(1), <1ms.
- Keyword match loop (responses list, 6 entries): O(k), <1ms.
- Chat history append: O(1) amortised list append, <0.1ms.
- Total end-to-end: well within 100ms SLA for personal use scale.

16. References

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